

CHAPTER 4: TABULATION PLAN MILDRED

Title: DETERMINANTS OF ANOGENITAL DISTANCE AND PENILE ANTHROPOMETRIC MEASURES IN MALE NEWBORNS: THE CASE OF TWO HOSPITALS IN BAMENDA

Objectives:

- To determine the normative value for anogenital distance and penile anthropometric measures in male newborn.
- To describe the variation of anogenital distance and penile anthropometric measures in male neonates of various ethnic groups.
- To identify the determinants of anogenital distance and penile anthropometric measures in male newborn babies

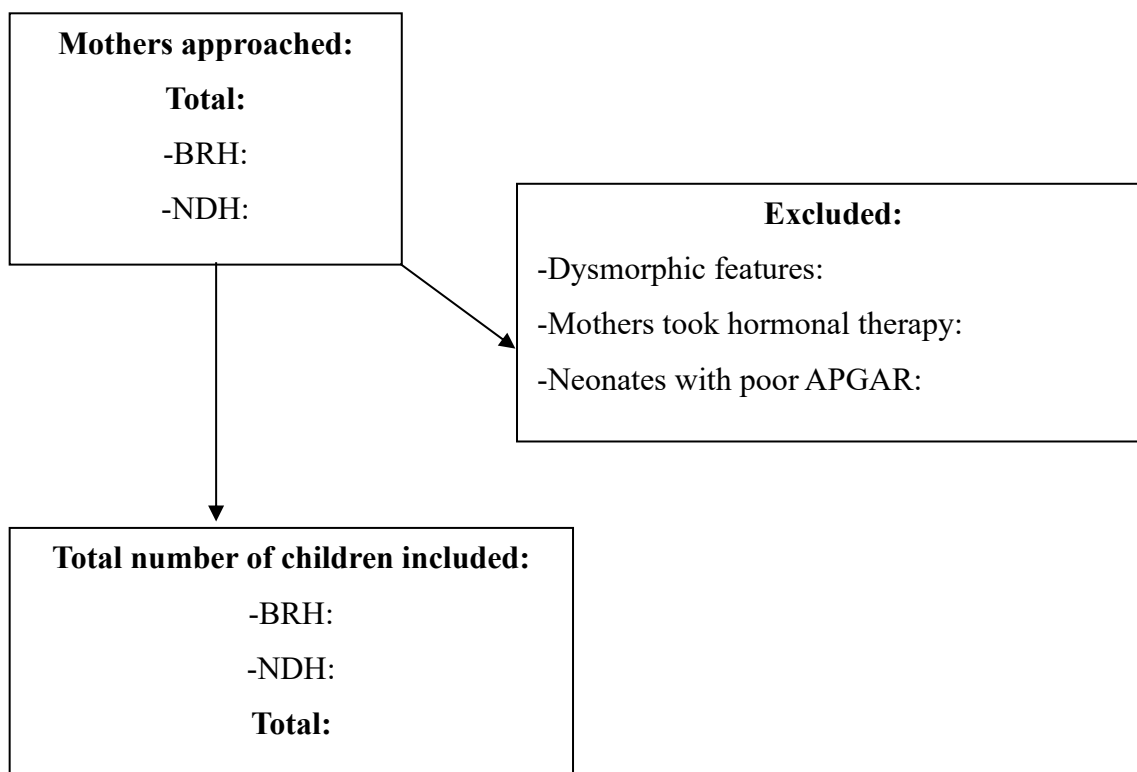


Figure 1: recruitment flow chart

4.1 DESCRIPTION OF THE STUDY POPULATION

4.1.1 Maternal sociodemographic characteristics

Table I: maternal sociodemographic characteristics of the study population

Characteristics	Number (N=)	Percentage %
Maternal age*	Mean $\pm$ SD	Min-max
Marital status		
Married/cohabiting		
Single/divorced/widowed		
Ethnic group		
Tikari		
Widikum		
Bamileke		
Duala		
Others		

*Mean $\pm$ standard deviation

4.1.2 Maternal obstetric history

Table II: maternal obstetric characteristics of our study population

Characteristics	Number (N=)	Percentage %
Gravidity		
Primigravida (1 pregnancy)		
Multigravida (2-4 pregnancies)		
Grand multigravida (≥ 5 pregnancies)		
Parity		
Primipara (1 delivery)		
Multipara (2-4 deliveries)		
Grand multipara (> 4 deliveries)		
History of PCOS		
Yes		
No		

4.1.3 maternal social history

Table III: maternal social history of the study population

Characteristics	Number (N=)	Percentage %
Passive smoking		
Yes		
No		
Active smoking		
Yes		
No		
Alcohol intake		
Yes		
No		
Drinking status		

Light drinker (AU <12.25 units)
Moderate drinker (AU 12.25 to 24.5 units)
Heavy drinker (AU >25.5 units)

4.1.4 maternal anthropometric measures

Table IV: Maternal anthropometric measures of the study population

Anthropometric measures	N (%)	Min-Max	Mean $\pm$ SD	Median (IQR)
Pre-pregnancy weight (kg)				
Height (m)				
BMI (kg/m²)				
Under weight				
Normal				
Overweight				
Grade I obesity				
Grade II Obesity				

SD: standard deviation; IQR: interquartile range

4.1.5 Infant characteristics

Table V: infants' characteristics of the study population

Characteristics	N (%)	Min-Max	Mean $\pm$ SD	Median (IQR)
GA at birth				
<37weeks				
37 – 40 weeks				
>40 weeks				
Mode of delivery				
Vaginal delivery				
Caesarian delivery				
Infant birth order				
Birth weight (g)				
Birth length (cm)				
Head circumference (cm)				

SD: standard deviation; IQR: interquartile range

4.2 NORMATIVE VALUES FOR GENITAL PARAMETERS

4.2.1 Distribution of genital parameters in our study population

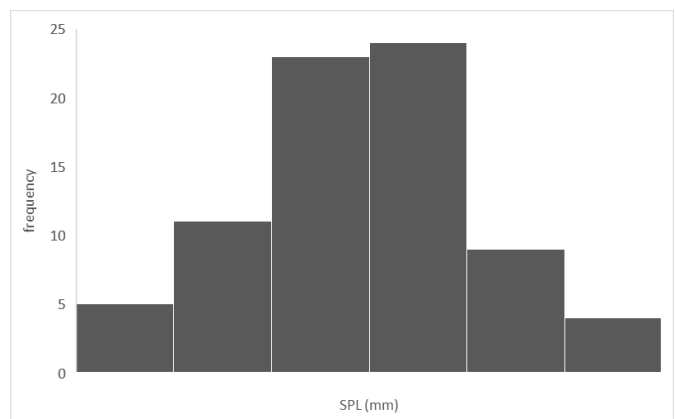
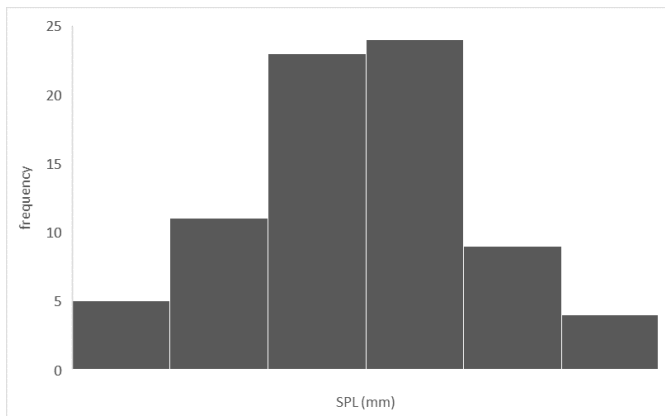
a) Summary statistics of genital parameters of male infants in the study population

Table VI: summary statistics of genital parameters of infants in the study population

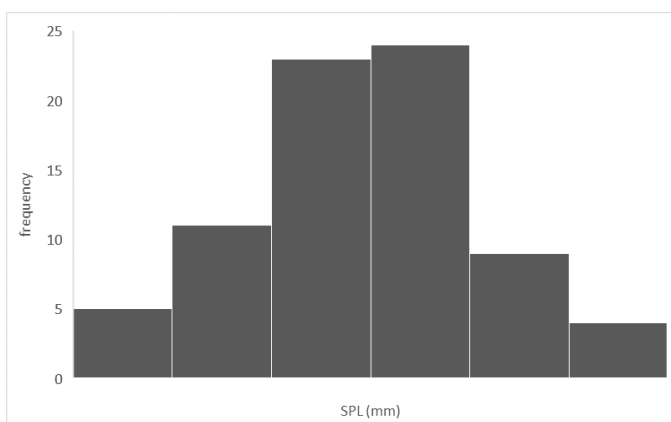
Genital parameter	N (%)	Min-Max	Mean $\pm$ SD	Median (IQR)
AGD-1				
AGD-2				
AGD-AS				
SPL				
PW				

SD: standard deviation; IQR: interquartile range

b) Distribution of anogenital distances in the study population (will be plotted on histograms)



c) Distribution of Peni dimensions in the study population (will be plotted on histograms)



4.2.4 Variation of genital parameters with gestational age at birth

NB: (Will be plotted on a line charts and side-by-side, showing the median and percentiles)

4.3 GENITAL PARAMETERS ACCORDING TO SUB-ETHNIC GROUPS

4.3.1 Variation of genital parameters in the different sub-ethnic groups

Table VII: distribution of genital parameters in the different sub-ethnic groups of the study population

Genital parameters	TIKAR (n)	WIDIKUM (n)	BAMILEKE (n)	CHAMBA (n)	TOTAL (n)
AGD-AS, mm					
p-value					
AGD-2, mm					
p-value					
AGD-1, mm					
p-value					
SPL, mm					
p-value					
PW, mm					
p-value					
Micropenis definition, mm					

*Comparison will be done among ethnic groups (between each other).

**Micropenis is defined as any penial length < mean SPL – 2.5SD

4.3.2 Distribution of SPL according to the different sub-ethnic groups

NB: Box plots will be drawn for each sub-ethnic group side-by-side to compare them.

4.4 DETERMINANTS OF GENITAL PARAMETERS

4.4.1 Factors associated or correlated to genital parameters

a) Maternal characteristics associated or correlated to genital parameters

Table VIII: maternal characteristics associated or correlated to genital parameters in the study population

Characteristics	AGD-1		AGD-2		AGD-AS	
	Mean ± SD	p-value (95% CI)	Mean ± SD	p-value (95% CI)	Mean ± SD	p-value (95% CI)
Maternal Age						
Gravidity (%)						
1						
2-4						
≥5						
Alcohol intake (%)						
No						
Yes						
Drinking status						
Light drinker						
Moderate drinker						
Heavy drinker						
Pre-pregnancy weight (kg)						
Height (m)						
BMI (kg/m)						

SD: standard deviation; CI: confidence interval

b) Maternal characteristics associated to penile dimensions

Table IX: maternal characteristics associated to penile dimensions in the study population

Characteristics	SPL		PW	
	Mean ± SD	p-value (95% CI)	Mean ± SD	p-value (95% CI)
Maternal Age				
Gravidity (%)				
1				
2-4				
≥5				
Alcohol intake (%)				
No				
Yes				
Drinking status				
Light drinker				
Moderate drinker				
Heavy drinker				
Pre-pregnancy weight (kg)				
Height (m)				
BMI (kg/m)				

SD: standard deviation; CI: confidence interval

c) Infant characteristics associated or correlated to anogenital dimensions

Table X: infant characteristics associated or correlated to anogenital dimensions in the study population

Characteristics	AGD-1		AGD-2		AGD-AS	
	Mean ± SD	p-value (95% CI)	Mean ± SD	p-value (95% CI)	Mean ± SD	p-value (95% CI)
GA at birth						
<37weeks						
37 – 40 weeks						
>40 weeks						
Mode of delivery						
Vaginal delivery						
Caesarian delivery						
Infant birth order						
Birth weight (g)						
Birth length (cm)						
Head circumference (cm)						

SD: standard deviation; CI: confidence interval

d) Infant characteristics associated or correlated to penile dimensions

Table XI: infant characteristics associated or correlated to penile dimensions in the study population

Characteristics	SPL		PW	
	Mean ± SD	p-value (95% CI)	Mean ± SD	p-value (95% CI)
GA at birth				
<37weeks				
37 – 40 weeks				
>40 weeks				
Mode of delivery				
Vaginal delivery				
Caesarian delivery				
Infant birth order				
Birth weight (g)				
Birth length (cm)				
Head circumference (cm)				

SD: standard deviation; CI: confidence interval

NB: Scatter plots with line smoother may be plotted to visualize the correlation between genital parameters and study characteristics that showed a significant correlation.

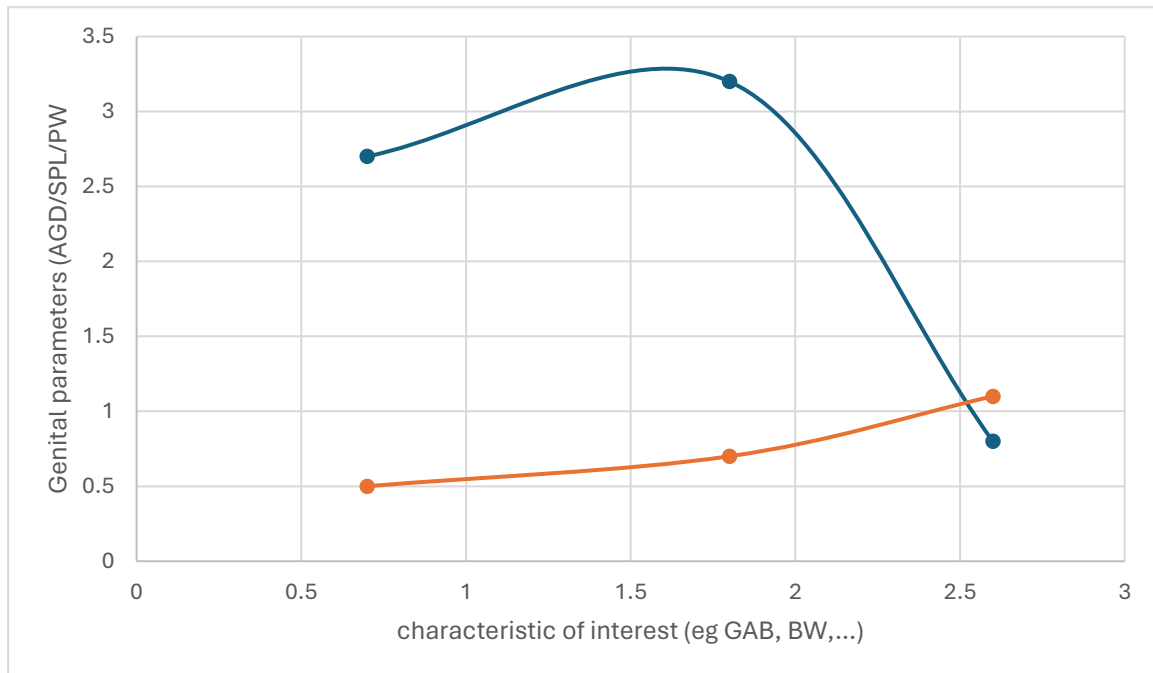


Figure 2: scatter plot showing the correlation between genital parameters and characteristics of interest

4.4.2 Predictors (determinants) of genital parameters

NB: Maternal and infant characteristics that will show a significant correlation or association with genital parameters will be fitted in a linear regression model to identify determinants of genital parameters while adjuster for confounders.

a) Determinants of anogenital distance

Table XII: determinants of anogenital distance of male infants in our study population

Characteristics	AGD-1		AGD-2		AGD-AS	
	Coefficient	p-value	Coefficient	p-value	Coefficient	p-value
Maternal characteristics	Maternal Age					
	Gravidity (%)					
	1					
	2-4					
	≥5					
	Alcohol intake (%)					
	No					
	Yes					
	Drinking status					
	Light drinker					
	Moderate drinker					
	Heavy drinker					
Infant characteristics	Pre-pregnancy weight (kg)					
	Height (m)					
	BMI (kg/m)					
	GA at birth					
	<37weeks					
	37 – 40 weeks					
	>40 weeks					
	Mode of delivery					
	Vaginal delivery					
	Caesarian delivery					
	Infant birth order					
	Birth weight (g)					
	Birth length (cm)					
	Head circumference (cm)					

b) Determinants of infants' penile dimensions

Table XIII: determinants of infants' penile dimensions in our study population

Characteristics	SPL		PW	
	Coefficient	p-value	Coefficient	p-value
Maternal characteristics	Maternal Age			
	Gravidity (%)			
	1			
	2-4			
	≥5			
	Alcohol intake (%)			
	No			
	Yes			
	Drinking status			
	Light drinker			
	Moderate drinker			
	Heavy drinker			
Infant characteristics	Pre-pregnancy weight (kg)			
	Height (m)			
	BMI (kg/m)			
	GA at birth			
	<37weeks			
	37 – 40 weeks			
	>40 weeks			
	Mode of delivery			
	Vaginal delivery			
	Caesarian delivery			
	Infant birth order			
	Birth weight (g)			
	Birth length (cm)			
	Head circumference (cm)			